

Sonu Yadav

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Professional Summary

Applied AI Engineer, and Data Scientist with full-stack AI/ML development experience from architecture to deployment. Proven expertise building production-grade LLM systems, RAG architectures, and predictive ML models that drive measurable business outcomes. Delivered 4+ AI products achieving 80% efficiency gains, 84.6% recall, and 70–85% cost reduction bridging technical innovation and domain-specific requirements.

Core Skills

AI/LLM: OpenAI API, LangChain, OpenAI Agent SDK, Function Calling, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Prompt Engineering, HuggingFace

Datastores: PostgreSQL, Redis, Pinecone, ChromaDB, Vector Database Design (chunking, metadata, hybrid search)

Backend/Platforms: FastAPI, Streamlit, Docker, Kubernetes, REST API Design, Pydantic

Data/ML: Python, scikit-learn, TensorFlow/Keras, Optuna, Ensemble Learning, Pandas, NumPy, Feature Engineering

Cloud/DevOps: AWS (S3, Lambda, ECS, ECR, CloudWatch, SageMaker), Git/GitHub, GitHub Actions (CI/CD), Docker, Kubernetes

Projects

HealthMate: Agentic Healthcare Assistant

LLM Engineer

- Built 8-microservice healthcare assistant with OpenAI Agent SDK, LangChain, Pinecone RAG, FastAPI, and Kubernetes; implemented JWT/OAuth2, RBAC, and HIPAA-compliant audit logging.
- Implemented AES-256 encryption and Prometheus/Grafana observability for secure, compliant patient data monitoring.

Stock Market Sentiment Analysis & Summarization (NLP, LLMs)

Data Scientist

- Delivered automated weekly market sentiment analysis and news summarization pipeline using Sentence Transformers and Mistral-7B, accelerating investment decision speed by 100x.
- Optimized quantized model deployment to produce >90% business-relevant summaries in JSON, enabling actionable insights for stakeholders.

ReneWind: Predictive Maintenance for Wind Turbines

Data Scientist

- Built 7-model ensemble achieving 84.6% recall on 20K+ samples with 17:1 imbalance, projecting 70–85% cost reduction.
- Delivered cost-benefit analysis and ROI recommendations to DOE stakeholders for proactive maintenance scheduling.

Restaurant Turnover Prediction — Hackathon Champion

Data Scientist

- Won 1st place with ensemble models (LightGBM, XGBoost, CatBoost) achieving 12.2M RMSE; engineered 30+ features.
- Proposed retention strategies that reduced projected turnover costs and improved workforce stability.

Professional Experience

Calitopia

Lead Programmer /Sr. Developer

San Francisco, CA

Aug 2025 – Present

- Delivered 3 B2B SaaS products to 5 prospective clients in 3 months with 100% milestones met, by managing a team of 10.
- Boosted marketing throughput 4x (2→8 weekly campaigns), by automating tasks in Python.
- Achieved zero bugs and 100% audit pass for AI sentiment tool, by modularizing code and fixing vulnerabilities.

Neurow

Technical Cofounder/AI Engineer

Austin, TX

Apr 2025 – Present

- Led end-to-end development of Neurow's AI powered productivity platform by architecting and deploying robust microservices.
- Implemented advanced data security protocols, achieving zero data breaches and ensuring HIPAA compliance.
- Collaborated with founders and engineers to build innovative features, including AI agents to prevent burnout and monitor mental health.

Natera Inc.

Clinical Laboratory Operator

Austin, TX

Feb 2023 – Nov 2024

- Processed 240–336 samples/day at 99% accuracy, optimizing workflows with QC analytics.
- Cut error rates by 95% through data-driven process improvements and standardized procedures.
- Standardized lab procedures, improving efficiency by 90% and enabling nationwide scalability.

University of Texas at Austin

Student Technological Assistant

Austin, TX

Jan 2022 – Dec 2022

- Cleaned and analyzed 10K+ records using SQL, Python, and Excel, and generated monthly reports improving system performance.
- Ensured reliable LMS operation for 1,000+ users, supporting data accuracy across 30+ courses.

Education

Westcliff University

Master's in Computer Science | GPA: 3.89

Dallas, TX

Aug 2026

University of Texas at Austin

Post-graduate Certification in AI/ML | GPA: 4.00

Austin, TX

Nov 2025

BS | Neuroscience | GPA: 3.63

Dec 2022

University of Texas at Arlington

BS | Biology | GPA: 3.63

Arlington, TX

Dec 2020

Certifications

- AI Agent Engineering – BreakIntoData
- Supervised Machine Learning – DeepLearning.AI (Coursera)
- Data Analytics Job Simulation – Accenture (Forage)